

Recognizer

A recognizer is a program that takes as input a string X and answers *yes* if X is a sentence of the language and *no* otherwise.

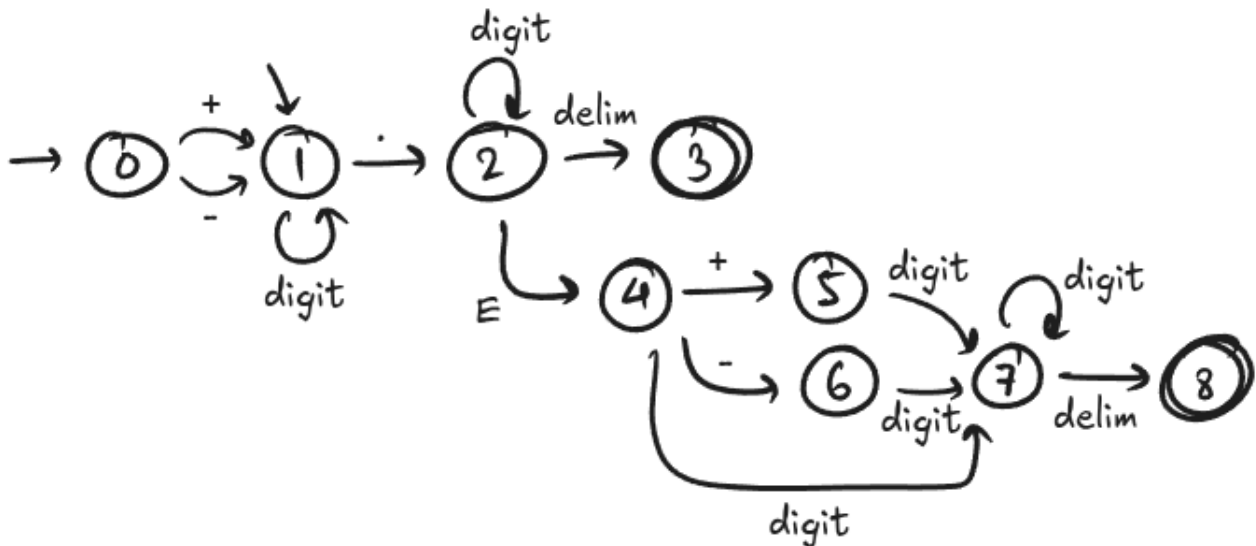


Figure 1: Recognizer for floating numbers

Conversion of NFA to DFA using Subset Construction Algorithm

Input: An NFA, N .

Output: A DFA, D accepting the same language.

Method:

- This algorithm constructs a transition table $Dtran$ for D .
- Each DFA state is a state of NFA state and we construct $Dtran$ which will simulate in parallel, all possible moves N can make on a given input string.

Notation:

- s is a NFA state
- T is a set of NFA states.

Operations:

- ϵ -closure(s): Set of NFA states reachable from NFA state s on ϵ transition alone.
- ϵ -closure(T): Set of NFA states reachable from some NFA states in T on ϵ transition alone. It is equal to the union of ϵ -closure(s) for each state s in T .
- move(T, a): Set of NFA states to which there is a transition on input symbol a from some NFA state s in T .

Algorithm:

Initially ϵ -closure(s_0) is the only state in D -states and it is unmarked.

while there is an unmarked state T in D -states do

begin

mark T

for each input symbol a do

begin

$U = \epsilon$ -closure(move(T, a))

if U is not in D -states then

add U as an unmarked state to D -states

$Dtran[T, a] = U$

end

end

To find ϵ -closure(T):

push all states of T into *stack*

initialize ϵ -closure(T) to T

while (*stack* is not empty) {

pop t off the *stack*

for each ϵ -transition u from t

if u is not in ϵ -closure(T)

push u into *stack*

add u to ϵ -closure(T)

}

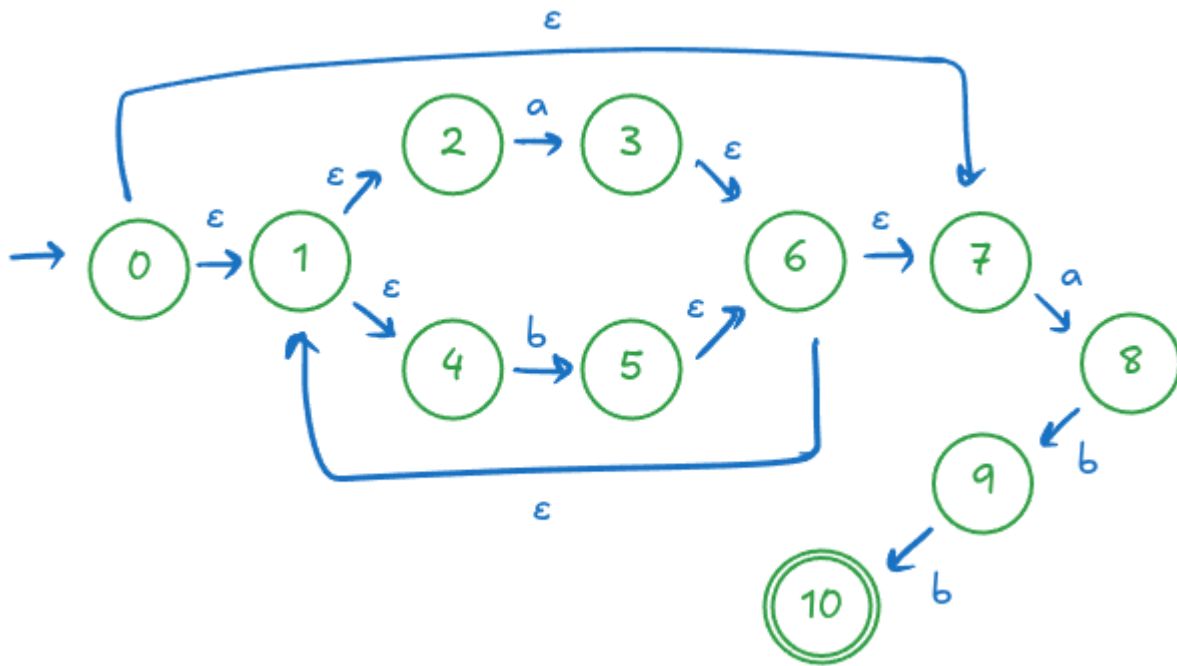


Figure 2: NFA accepting $(a/b)^*abb$

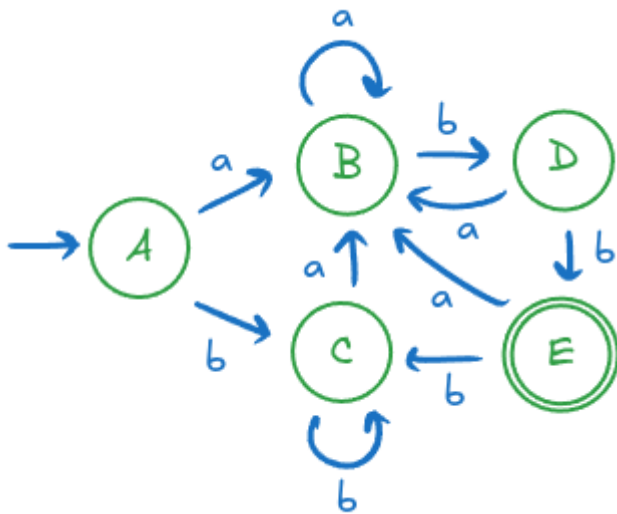


Figure 3: DFA constructed using subset construction method

Operation	D-states	a	b
$\epsilon\text{-closure}(0)$	$\{0, 1, 2, 4, 7\} = A$	B	c
$\epsilon\text{-closure}(\text{move}(A, a))$ $= \epsilon\text{-closure}(\{3, 8\})$	$\{1, 2, 3, 4, 6, 7, 8\} = B$	B	D
$\epsilon\text{-closure}(\text{move}(A, b))$ $= \epsilon\text{-closure}(\{5\})$	$\{1, 2, 4, 5, 6, 7\} = C$	B	c
$\epsilon\text{-closure}(\text{move}(B, a))$ $= \epsilon\text{-closure}(\{3, 8\}) = B$			
$\epsilon\text{-closure}(\text{move}(B, b))$ $= \epsilon\text{-closure}(\{5, 9\})$	$\{1, 2, 4, 5, 6, 7, 9\} = D$	B	E
$\epsilon\text{-closure}(\text{move}(C, a))$ $= \epsilon\text{-closure}(\{3, 8\}) = B$			
$\epsilon\text{-closure}(\text{move}(C, b))$ $= \epsilon\text{-closure}(\{5\}) = C$			
$\epsilon\text{-closure}(\text{move}(D, a))$ $= \epsilon\text{-closure}(\{3, 8\}) = B$			
$\epsilon\text{-closure}(\text{move}(D, b))$ $= \epsilon\text{-closure}(\{5, 10\})$	$\{1, 2, 4, 5, 6, 7, 10\} = E$	B	c
$\epsilon\text{-closure}(\text{move}(E, a))$ $= \epsilon\text{-closure}(\{3, 8\}) = B$			
$\epsilon\text{-closure}(\text{move}(E, b))$ $= \epsilon\text{-closure}(\{5\}) = C$			

Table 1: Dtran table used to convert DFA in Figure 3